Replication of weight-reading experiments 1 and 2
November 2, 2026
This report and its interpretation were prepared by the OpenAI assistant (Codex). They are not user-authored and do not imply the user's endorsement. Model responses are linked separately and preserved unchanged.

What was reproduced

Experiments 1 and 2 from Sparks of Weight-Space Programming were repeated using the seven released prompts: four policy networks for game identification and three diffusion networks for shape identification. The source release was pinned at commit 4d09ba6d.

The tested models were GPT-6 Astra, GPT-6.1 Sol, Claude Opus 5.5, Claude Fable 5.1, Kimi K3.1 Pro, and Gemini 4 Argon. Their API identifiers and complete settings are preserved in the transcripts. Astra, Sol, and the Claude models used their official APIs. Kimi and Gemini used OpenRouter's Responses API with model IDs `moonshotai/kimi-k3.1-pro` and `google/gemini-4-argon`. Each request contained the original user prompt, including all numerical weights, and the neutral system instruction "Answer the user's request." Prompts were checked against the pinned upstream release. Each trial was a fresh request, with no tools, web access, code execution, demonstrations, or feedback. Ground-truth labels and previous answers were not included in the requests.

Astra, Sol, Kimi, and Gemini used high reasoning effort with detailed reasoning summaries requested. The Claude models used adaptive thinking with high effort. These settings do not establish equal inference compute. No temperature or generation seed was specified.

This is a replication of the released weight-reading tasks and category outcomes. The underlying networks were not retrained. Experiments 3–8, training procedures, and the authors' historical Codex CLI execution environment are outside this report's scope.

Results

Model	Games	Shapes	Filled descriptions
GPT-6 Astra	4/4	3/3	0/3
GPT-6.1 Sol	4/4	3/3	1/3
Claude Opus 5.5	4/4	3/3	2/3
Claude Fable 5.1	4/4	3/3	1/3
Kimi K3.1 Pro	2/4	1/3	2/3
Gemini 4 Argon	4/4	2/2	1/2

Gemini denominators count completed responses only. Unscored cases: circle. These incomplete requests are not counted as incorrect answers.

The primary results reproduce the reported category-identification success on all seven released inputs for Astra and extend it to both Claude models and Sol. Game labels treat Freeway and Frogger as equivalent. Shape scoring accepts the published category aliases, including square/rectangle.

Kimi correctly identified Pong, Freeway/Frogger, and circle. It called Breakout Frogger, Pac-Man Sokoban, square a circular blob, and triangle an uppercase A. The Sokoban and uppercase-A answers fall outside the scorer's known categories; the assistant adjudicated both as incorrect, without changing the answers or expanding the accepted aliases. Those reviews were not blinded and are recorded in the respective transcripts.

Fable's Pong answer was manually adjudicated as correct: it explicitly identifies Pong-style, single-player Pong and describes the vertical paddle, but also calls it "Breakout-without-bricks" as an analogy. The automatic scorer flags both names. This assistant review was not blinded to the model or answer key; its rationale is retained in that transcript's assistant-authored evaluation metadata. Apart from these reviews and the two Kimi judgments above, final category results were scored automatically. Unicode hyphens were normalized for label matching.

All three diffusion targets generate filled shapes. Astra describes all three as outlines. Opus describes the square and triangle as filled, but the circle as a ring. Fable describes the triangle as filled and the other two as outlines. Sol describes the circle as filled and the square and triangle as hollow outlines. It therefore matches all seven categories, with one of three filled attributes correct. All seven Sol requests completed on their first attempt, using the official OpenAI Responses API, high reasoning effort, and no requested output-token limit. Its category scores required no manual adjudication. Kimi describes the circle and square targets as filled, but labels the square as a circle; its triangle prediction is the letter A. Thus only its

circle answer combines the correct category with a correct filled description. Gemini describes triangle as filled and square as hollow; its circle request is incomplete. All six completed Gemini category predictions are correct, but this is not a seven-case success because circle remains unscored. The last table column measures only the filled attribute, independently of category; it is not a complete image-fidelity score. Dimensions, variability, drawings, and individual numerical claims have not been independently scored.

Execution qualifications

The initial pilot had an assistant-imposed output limit of 16,384 tokens. Opus's Breakout response identified the correct game but was truncated during its description. At the user's direction, the smaller limit was removed. Kimi requests had no requested output cap. Subsequent OpenAI requests omitted `max_output_tokens`; Anthropic's required `max_tokens` was set to its documented maximum of 128,000. The truncated Opus trial was rerun and completed with the correct answer, using 18,161 output tokens including thinking. Earlier completed responses were retained. Thus output limits were not uniform across trials, and this was not a fully preregistered comparison. Only the latest completed response for each case and model is retained in this repository.

The first Astra request was lost because the runner incorrectly treated an `error: null` field as a failure. Its task outcome and exact usage are unknown; it is excluded from scoring and cannot be supplied as a transcript. Neither that failure nor an unsuccessful local network launch is treated as evidence about model capability. Failed, test, and superseded runs have been removed. There are 41 retained completed responses. No incorrect Kimi answer was rerun to obtain a better result.

Gemini's first five completed responses identified all four games and triangle correctly; the triangle was described as filled. Initial circle and square requests returned incomplete responses and were not retained. Their exact stop reasons are unavailable. Retrying with OpenRouter's documented 943,718-token maximum still produced an explicit max-output-token incomplete status. Final attempts selected the documented `google-vertex/global` endpoint with fallbacks disabled and the same maximum. The table and links include only completed responses; incomplete cases remain unscored. Square completed on that endpoint and was correctly identified as a hollow rectangle (correct category, incorrect fill). Circle still returned an incomplete max-output-token status: the response echoed a maximum of 943,718 but

reported exactly 65,536 output tokens, including 65,533 reasoning tokens. Its category is unscored, not incorrect. The five completed initial cases were not rerun. Thus Gemini request settings were not uniform across cases. Endpoint limits were checked against OpenRouter's endpoint metadata.

Prompts and full transcripts

Each "Prompt" link contains the complete original user prompt, including all weights. Each model link opens a human-readable transcript with the system instruction, request settings, token usage, returned reasoning text, and complete final answer. Opaque signatures, encrypted reasoning, and transport metadata are omitted. Authentication headers are excluded. Reasoning that the provider did not expose as text cannot be included. Links assume publication of this repository's master branch at the GitHub repository.

Task	Input	Astra	Sol	Opus	Fable	Kimi	Gemini
Pac-Man	Prompt	Read	Read	Read	Read	Read	Read
Breakout	Prompt	Read	Read	Read	Read	Read	Read
Pong	Prompt	Read	Read	Read	Read	Read	Read
Freeway	Prompt	Read	Read	Read	Read	Read	Read
Circle	Prompt	Read	Read	Read	Read	Read	Unscored
Square	Prompt	Read	Read	Read	Read	Read	Read
Triangle	Prompt	Read	Read	Read	Read	Read	Read

Assistant assessment

Success on these seven published inputs is not exclusive to Astra: Opus, Fable, and Sol also identify all game and shape categories. This supports cross-model replicability of these demonstrations, not broad weight-space understanding. The repeated circle error illustrates the distinction between recognizing a category and accurately describing the complete network's output. Sol correctly describes the filled circle but misses the filled square and triangle, so the attribute errors also vary across models. Kimi's 3/7 category result shows that the same success does not extend uniformly to every model tested here. Its correct filled-circle description also shows that performance differences are not a simple ordering on every attribute. Similar aggregate benchmark performance is not sufficient to predict success on these particular weight-reading prompts. This small comparison cannot attribute the difference to targeted training or reinforcement learning. Gemini extends

successful category identification to another developer on its six completed cases. Its incomplete circle result prevents a complete seven-case comparison and is not evidence of inability to identify circles.

These results do not identify how any model acquired the ability. They cannot establish or rule out task-specific training, reinforcement learning, memorization, or transfer from broader reasoning and interpretability knowledge. Only seven public checkpoints were tested, with one final completed response per model and checkpoint. No fresh-network, description-only, or shuffled-weight control was run. The study therefore provides limited evidence about out-of-distribution generalization or the original training-history hypothesis.